Attendance Recovery Computation

Problem: Calculate the current attendance percentage and determine the minimum consecutive classes a student must attend to reach the required eligibility percentage under different future attendance scenarios.

1. Computational Thinking

1. Decomposition:

- Input total classes conducted.
- Input classes attended.
- Calculate current attendance percentage.
- Check eligibility percentage.
- Calculate minimum additional classes required.
- Display the final attendance status.

2. Pattern Recognition:

- Attendance increases when a student attends future classes.
- More classes are required when current attendance is lower.
- Different future attendance scenarios produce different results.

3. Abstraction:

- Focus on total classes, attended classes and eligibility percentage.
- Ignore unnecessary student details.

4. Algorithm Design:

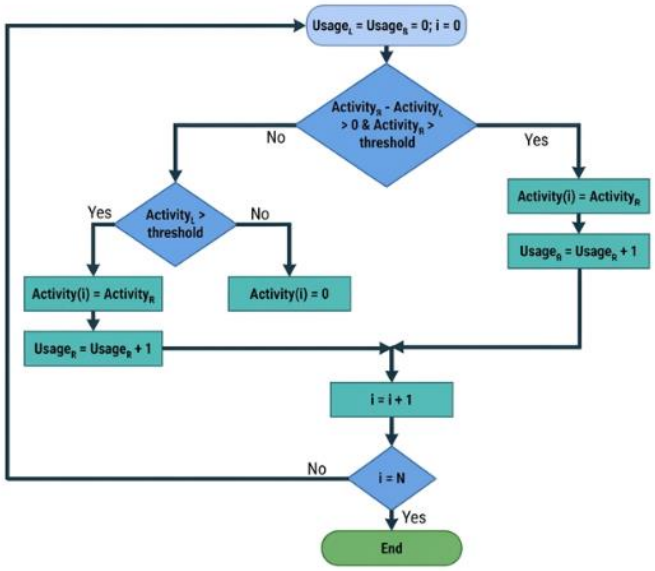
- Use formulas, loops and conditional statements to calculate attendance recovery.

2. Algorithm

1. Start.
2. Input student name.
3. Input total classes conducted.
4. Input classes attended.
5. Input required eligibility percentage.
6. Validate the input values.
7. Calculate current attendance percentage.

8. Check whether the student is already eligible.
9. If not eligible, initialize additional classes to zero.
10. Use a loop to calculate future attendance.
11. Increase attended classes and total classes for each future class attended.
12. Repeat until the required percentage is reached.
13. Display minimum additional classes required.
14. Calculate and display final attendance percentage.
15. Stop.

3. Flowchart:



4. Variables

Variable	Purpose
name	Student name
total	Total classes conducted
attended	Classes attended
percentage	Current attendance
required	Eligibility percentage

extra	Additional classes required
future	Future classes
final_attendance	Final attendance percentage
i	Loop counter

5. Data Types

Data type	Example
Integer	100, 75, 10
Float	75.5, 82.5
String	Student name
Boolean	True / False
List	Future attendance scenarios
Tuple	(total, attended)

6. Operators Used

Operator	Purpose
+	Add attended classes
-	Calculate absent classes
*	Calculate percentage
/	Divide attendance
>=	Check eligibility
==	Compare values
+=	Increase additional classes
and	Validate multiple conditions

Attendance formula:

$$ \text{Attendance \%} = \frac{\text{Classes Attended}}{\text{Total Classes}} \times 100 $$

7. Conditional Statements

- if: Check whether attendance is already eligible.
- else: Calculate additional classes if not eligible.
- if: Validate total and attended classes.
- while: Continue until eligibility is reached.

8. Loops

- While loop: Calculate the minimum consecutive classes required.
- For loop: Process different future attendance scenarios.

9. Strings

Strings store student names and attendance status.

name = "Rahul"status = "Eligible"

10. Pattern Generation

Generate an attendance report.

ATTENDANCE REPORT

Student Name: Rahul

Total Classes: 100

Attended Classes: 70

Current Attendance: 70%

Required Attendance: 75%

Extra Classes Required: 20

Final Attendance: 75%

Status: ELIGIBLE

11. Lists

Lists store different future attendance scenarios.

scenarios = [1, 2, 3, 4, 5]

12. Tuples and Packing

A tuple stores total classes and attended classes together.

Packing:

attendance = (100, 70)

Unpacking:

total, attended = attendance

13. Pseudocode

START

INPUT student_name

INPUT total_classes

INPUT attended_classes

INPUT required_percentage

IF total_classes <= 0 OR

  attended_classes < 0 OR

  attended_classes > total_classes THEN

   DISPLAY "Invalid Input"

   STOP

END IF

SET percentage =

   (attended_classes / total_classes) * 100

DISPLAY percentage

IF percentage >= required_percentage THEN

   DISPLAY "Already Eligible"

```
    extra = 0

ELSE

    SET extra = 0

    WHILE TRUE

        extra = extra + 1

        new_total = total_classes + extra
        new_attended = attended_classes + extra

        new_percentage =
            (new_attended / new_total) * 100

        IF new_percentage >= required_percentage THEN
            BREAK
        END IF

    END WHILE

    DISPLAY extra

END IF

DISPLAY "Final Attendance Percentage"
DISPLAY new_percentage
```

STOP

14. Python Program

```
name = "Rahul"total = 100attended = 70required = 75percentage = (attended / total) * 100print("Current Attendance:", percentage, "%")if total <= 0 or attended < 0 or attended > total:    print("Invalid Input")elif percentage >= required:    print("Already Eligible")    print("Extra Classes Required: 0")else:    extra = 0    while True:        extra += 1        new_total = total + extra        new_attended = attended + extra        final_percentage = (new_attended / new_total) * 100        if final_percentage >= required:            break    print("Extra Classes Required:", extra)    print("Final Attendance:", round(final_percentage, 2), "%")    print("Status: Eligible")
```

15. Expected Output

Current Attendance: 70.0 %

Extra Classes Required: 20

Final Attendance: 75.0 %

Status: Eligible

16. Different Future Scenarios

Suppose a student has attended 70 out of 100 classes and needs 75% attendance.

Future scenario	**Attendance calculation**
Attend every future class	(70+x)/(100+x) × 100
Attend 80% of future classes	(70+0.8x)/(100+x) × 100
Attend 50% of future classes	(70+0.5x)/(100+x) × 100

Important: If the student attends less than the required 75% of future classes, reaching 75% overall may not be possible.

17. Conclusion

Attendance Recovery Computation uses computational thinking, algorithms, flowcharts, variables, data types, operators, conditional statements, loops, strings, lists and tuples to calculate current attendance and determine the minimum additional classes needed to achieve eligibility under different future attendance scenarios.